

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Debugging & Optimization Task

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	Total Marks:	100
CO2 Statement:	Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.		
Student Name:	SANKAR R	Reg. No.:	192525131
Section / Batch:	SLOT-B	Date:	22/07/2026

Code of Conduct

I am SANKAR R(192525131) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: SANKAR R

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure	List Operations	6 – 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
GRAND TOTAL:		100 Marks	

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, well-documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well-justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

[← Back](#)

QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack
5 marks

Q6

Q8 Linked List Data Structures

5 marks

Identify the errors in the following code

```
while current != NULL:
    current.next = prev
    prev = current
    current = current.next
```

 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 while (current != NULL) {
2     next_node = current.next; // save next
3     current.next = prev;      // reverse link
4     prev = current;           // move prev forward
5     current = next_node;      // move current forward
6 }
```

Test Input

stdin input (optional)

Output

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

[← Back](#)

QUESTIONS

Q1Binary Search
5 marks**Q2**Stack
5 marks**Q3**Stack
5 marks**Q4**Stack
5 marks**Q5**Stack
5 marks**Q6**Linked List
5 marks

Q10 Queue

5 marks

What is missing in the below pseudocode?

DEQUEUE(Q):

if front == -1:

print("Underflow")

else:

return Q[front++]

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 DEQUEUE(Q) :  
2     if front == -1 OR front > rear:  
3         print("Underflow")  
4     else:  
5         return Q[front++]
```

Test Input

stdin input (optional)

Output



QUESTIONS

Q1Stack Notations - Postfix
Expression
5 marks**Q2**Circular Queue
5 marks**Q3**Binary Search
5 marks**Q4**Queue
5 marks**Q5**Queue
5 marks**Q6**Doubly Linked List
5 marks**Q7**Stack
5 marks

Q10 Stack

5 marks

Point out the error?

PEEK(stack):

return stack[top+1]

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 PEEK(stack) :  
2     return stack[top]
```

Test Input

stdin input (optional)

Output



Run



Save



Submit Fix

QUESTIONS

Q1
Stack Notations - Postfix Expression
5 marks

Q2
Circular Queue
5 marks

Q3
Binary Search
5 marks

Q4
Queue
5 marks

Q5
Queue
5 marks

Q6
Doubly Linked List
5 marks

Q7
Stack
5 marks

Q9 Circular Queue

5 marks

Identify the issue in the following snippet for Circular Queue in Traffic Signal.

```
rear = (rear + 1) % MAX;
queue[rear] = vehicle;
if(rear == front){
printf("Queue Full");
}
```

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 // Check if the queue is full before inserting
2 if ((rear + 1) % MAX == front) {
3     printf("Queue Full\n");
4 } else {
5     rear = (rear + 1) % MAX;
6     queue[rear] = vehicle;
7 }
```

Test Input

stdin input (optional)

Output
QUESTIONS

Q1
Stack Notations - Postfix Expression
5 marks

Q2
Circular Queue
5 marks

Q3
Binary Search
5 marks

Q4
Queue
5 marks

Q5
Queue
5 marks

Q6
Doubly Linked List
5 marks

Q7
Stack
5 marks

Q8
Stack

Q8 Stack

5 marks

Point out the error.
if top == MAX:
Overflow

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 if top == MAX - 1:
2     print("Overflow")
```

Test Input

stdin input (optional)

Output
 Run

 Save

 Submit Fix

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack
5 marksQ6
Linked List
5 marksQ7
Stack
5 marks

Q2 Stack

5 marks

Find the error in the stack deletion operation. Justify it.

```
int pop() {  
    if(top == 0) {  
        printf("Underflow");  
        return -1;  
    }  
    return stack[top--];  
}
```

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

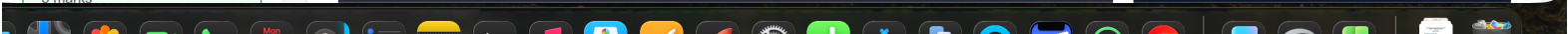
Your Fixed Code

```
1 int pop() {  
2     if(top == -1) {  
3         printf("Underflow");  
4         return -1;  
5     }  
6     return stack[top--];  
7 }
```

Test Input

stdin input (optional)

Output



CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

[← Back](#)

QUESTIONS

- Q1

Binary Search

5 marks

✓
- Q2

Stack

5 marks

✓
- Q3

Stack

5 marks

✓
- Q4

Stack

5 marks

✓
- Q5

Stack

5 marks

✓
- Q6

Linked List

5 marks

✓
- Q7

Stack

5 marks

✓

Q3 Stack

5 marks

Check the snippet and identify the logical error.

```
int stack[5];
int top;
void init() {
    top = 0;
}
```

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #define max 5
3 int stack[max];
4 int top;
5 void init()
6 {
7     top = -1;
8 }
9 int main ()
10 {
11     init();
```

Test Input

stdin input (optional)

Output

Top=-1

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1
Stack Notations - Postfix Expression
5 marks

Q2
Circular Queue
5 marks

Q3
Binary Search
5 marks

Q4
Queue
5 marks

Q5
Queue
5 marks

Q6
Doubly Linked List
5 marks

Q7

Q1 Stack Notations - Postfix Expression 5 marks

Find the bug in the evaluation of a Postfix expression.

```
while (expression[i] != '\0') {  
    if (isdigit(expression[i])) push(expression[i] - '0');  
    else {  
        int op1 = pop();  
        int op2 = pop();  
        // perform operation: result = op1 + op2;  
        push(result);  
    }  
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int i = 0;  
2 int result;  
3 while (expression[i] != '\0') {  
4     if (isdigit(expression[i])) {  
5         push(expression[i] - '0');  
6     } else {
```

Test Input
stdin input (optional)

Output

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1
Stack Notations - Postfix Expression
5 marks

Q2
Circular Queue
5 marks

Q3
Binary Search
5 marks

Q4
Queue
5 marks

Q5
Queue
5 marks

Q6
Doubly Linked List
5 marks

Q2 Circular Queue 5 marks

Identify the bug in this Circular Queue dequeue logic.

```
int dequeue() {  
    if (front == -1) return -1;  
    int data = queue[front];  
    front = (front + 1) % MAX;  
    return data;  
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int dequeue() {  
2     if (front == -1) {  
3         return -1;  
4     }  
5  
6     int data = queue[front];  
7
```

Test Input
stdin input (optional)

Output

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1
Stack Notations - Postfix
Expression
5 marksQ2
Circular Queue
5 marksQ3
Binary Search
5 marksQ4
Queue
5 marksQ5
Queue
5 marksQ6
Doubly Linked List
5 marks

Q7

Q3 Binary Search

5 marks

Identify the bug in this Recursive Binary Search.

```
int binarySearch(int arr[], int l, int r, int x) {  
    int mid = l + (r - l) / 2;  
    if (arr[mid] == x) return mid;  
    if (arr[mid] > x) return binarySearch(arr, l, mid, x);  
    return binarySearch(arr, mid, r, x);  
}
```

🔗 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int binarySearch(int arr[], int l, int r, int x) {  
2     if (l > r) return -1;  
3  
4     int mid = l + (r - l) / 2;  
5  
6     if (arr[mid] == x) return mid;  
7  
8     if (arr[mid] > x) return binarySearch(arr, l, mid - 1, x);  
9     return binarySearch(arr, mid + 1, r, x);  
}
```

Test Input

stdin input (optional)

Output



Video for Sankar R

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack
5 marksQ6
Linked List
5 marksQ7
Stack
5 marks

Q4 Stack

5 marks

Identify the errors in the following snippet.

```
if(ch == ')') {  
    while(stack[top] != ')') {  
        postfix[j++] = pop();  
    }  
    pop();  
}
```

🔗 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 char stack[100];  
3 char postfix[100];  
4 int top = -1, j = 0;  
5 char ch;  
6  
7 char pop() {  
8     {  
9         return stack[top--];  
10 }
```

Test Input

(a+b)*c

Output

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1
Stack Notations - Postfix
Expression
5 marksQ2
Circular Queue
5 marksQ3
Binary Search
5 marksQ4
Queue
5 marksQ5
Queue
5 marksQ6
Doubly Linked List
5 marksQ7
Stack

Q4 Queue

5 marks

Find the error in this Deque insertFront operation.

```
void insertFront(int data) {  
    if (front == 0) front = MAX - 1;  
    else front = front - 1;  
    deque[front] = data;  
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 void insertFront(int data) {  
2     if (isFull()) return; // or handle overflow  
3     if (front == -1) { // empty deque  
4         front = rear = 0;  
5         deque[front] = data;  
6     } else {  
7         front = (front == 0) ? (MAX - 1) : (front - 1);  
8         deque[front] = data;  
9     }  
10 }
```

Test Input

stdin input (optional)

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack
5 marksQ6
Linked List
5 marksQ7
Stack

Q5 Stack

5 marks

In balancing Bracket, you identify the error.

```
if((ch == '[' && stack[top] == '[') ||  
(ch == ']' && stack[top] == '[')) {  
    pop();  
} else {  
    return 0;  
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 if (top == -1) {  
2     return 0; // nothing to match with, unbalanced  
3 } else if ((ch == '[' && stack[top] == '[') ||  
4           (ch == ']' && stack[top] == '[') ||  
5           (ch == '[' && stack[top] == '[') {  
6     pop();  
7 } else {  
8     return 0;  
9 }
```

Test Input

stdin input (optional)

Output

QUESTIONS

- Q1

Stack Notations - Postfix Expression

5 marks
- Q2

Circular Queue

5 marks
- Q3

Binary Search

5 marks
- Q4

Queue

5 marks
- Q5

Queue

5 marks
- Q6

Doubly Linked List

5 marks
- Q7

Stack

5 marks

Q5 Queue

5 marks

Identify the error in the Queue enqueue logic.

```
void enqueue(int data) {  
    if (rear == MAX) return;  
    queue[rear++] = data;  
}
```

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 void enqueue(int data) {  
2     if (rear == MAX - 1) {  
3         return;  
4     }  
5     queue[++rear] = data;  
6 }
```

Test Input

stdin input (optional)

Output

QUESTIONS

- Q1

Binary Search

5 marks
- Q2

Stack

5 marks
- Q3

Stack

5 marks
- Q4

Stack

5 marks
- Q5

Stack

5 marks
- Q6

Linked List

5 marks
- Q7

Stack

5 marks

Q6 Linked List

5 marks

What is the issue?

```
temp = head  
while temp.next != NULL:  
    print(temp.data)  
    temp = temp.next
```

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 temp = head  
2 while temp != None:  
3     print(temp.data)  
4     temp = temp.next
```

Test Input

stdin input (optional)

Output

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

[← Back](#)

QUESTIONS

Q1 ✓
Stack Notations - Postfix Expression
5 marks

Q2 ✓
Circular Queue
5 marks

Q3 ✓
Binary Search
5 marks

Q4 ✓
Queue
5 marks

Q5 ✓
Queue
5 marks

Q6 ✓
Doubly Linked List
5 marks

Q7 ✓
Stack

Q6 Doubly Linked List

5 marks

Identify the error in this Doubly Linked List node deletion.

```
void deleteNode(Node* del) {
    if (del->prev != NULL)
        del->prev->next = del->next;
    if (del->next != NULL)
        del->next->prev = del->prev;
}
```

 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 void deleteNode(Node* del) {
2     if (del == nullptr) return;
3
4     // Update previous node's next pointer
5     if (del->prev != nullptr)
6         del->prev->next = del->next;
7
8     // Update next node's prev pointer
9     if (del->next != nullptr)
10        del->next->prev = del->prev;
```

Test Input

stdin input (optional)

Output

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

[← Back](#)

QUESTIONS

Q1
Binary Search
5 marks**Q2**
Stack
5 marks**Q3**
Stack
5 marks**Q4**
Stack
5 marks**Q5**
Stack
5 marks**Q6**
Linked List
5 marks**Q7**
Stack**Q7 Stack**

5 marks

Point out the error.

POP(stack):

if top == 0:

print("Underflow")

else:

return stack[top--]

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 def POP(stack):  
2     global top  
3     if top == -1:  
4         print("Underflow")  
5     else:  
6         value = stack[top]  
7         top = top - 1  
8         return value
```

Test Input

stdin input (optional)

Output

Q1
Binary Search
5 marks



Q2
Stack
5 marks



Q3
Stack
5 marks



Q4
Stack
5 marks



Q5
Stack
5 marks



Q6
Linked List
5 marks



Q7
Stack
5 marks



Q8
Linked List Data Structures
5 marks



Identify the error in the following snippet.

```
int binarySearch(int arr[], int n, int key) {  
    int low = 0, high = n - 1;  
    while(low < high) {  
        int mid = (low + high) / 2;  
        if(arr[mid] == key)  
            return mid;  
        else if(arr[mid] < key)  
            low = mid;  
        else  
            high = mid;  
    }  
    return -1;  
}
```

 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 int binarysearch(int arr[],int n,int key){  
3     int low = 0,high = n-1;  
4     while (low<=high){  
5         int mid = low + (high - low)/2;  
6         if (arr[mid]==key)  
7             return mid;
```

Test Input

stdin input (optional)

Output

QUESTIONS

Q1
Stack Notations - Postfix
Expression
5 marks

Q2
Circular Queue
5 marks

Q3
Binary Search
5 marks

Q4
Queue
5 marks

Q5
Queue
5 marks

Q6
Doubly Linked List
5 marks

Q7
Stack
5 marks

Q8

Q7 Stack

5 marks

Identify the issue in the below pseudocode.
POP():
if top == 0 → Underflow

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 POP() :
2     if top == -1:
3         print("Underflow")
4     else:
5         top = top - 1
```

Test Input

stdin input (optional)

Output

```
Compilation Error:
C:\Windows\TEMP\sandbox_192525131
RB_6a794d403ad8e8.28497442\main.c
:1:1: warning: return type
defaults to 'int' [-Wimplicit-
int]
```

 Run

 Save

 Submit Fix